গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
Six Storied Residential Building

AT PLOT # 27, BLOC-B, ROAD # 22, SHAHJALAL UPASHAHOR, SYLHET
DAG # SA-225, BS-4454, KHA # SA-563, BS-4438, NAMJARI-3724
J.L# SA-98, BS-79, MOUJA- SADIPUR 2ND PART

CLIENT

NAGIB MAHFUZ CHOWDURY NAFIS

Verification Code: 53g0Zfhc
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-50 ft	10
02	Disturbed sample	02	0-50 ft	10
03	Disturbed sample	03	0-50 ft	10
04	Disturbed sample	04	0-50 ft	10

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$ (Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_u = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = 4N/200\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square (Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	6	0.696	0.732	0.798	0.822	0.888	0.966	0.750
	5.0-10	11	1.220	1.283	1.399	1.441	1.557	1.693	1.315
BH02	2.5-5.0	7	0.835	0.878	0.958	0.986	1.066	1.159	0.900
	5.0-10	11	1.244	1.308	1.426	1.469	1.587	1.727	1.341
BH03	2.5-5.0	6	0.673	0.708	0.771	0.795	0.858	0.934	0.725
	5.0-10	10	1.176	1.237	1.348	1.389	1.500	1.632	1.267
BH04	2.5-5.0	3	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	8	0.952	1.001	1.092	1.124	1.215	1.321	1.026
BH05	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

** Designer should consider both the calculated design N value and settlement behavior of the corresponding soil profile while selecting shallow footings

Table 8: The allowable skin friction for precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.096	0.048	0.074	0.096	-	-	-	-	-	-
10	0.118	0.072	0.138	0.049	-	-	-	-	-	-
15	0.174	0.048	0.156	0.096	-	-	-	-	-	-
20	0.144	0.136	0.136	0.144	-	-	-	-	-	-
25	0.118	0.096	0.096	0.096	-	-	-	-	-	-
30	0.160	0.160	0.152	0.144	-	-	-	-	-	-
35	0.168	0.168	0.160	0.112	-	-	-	-	-	-
40	0.168	0.168	0.160	0.112	-	-	-	-	-	-
45	0.168	0.176	0.160	0.160	-	-	-	-	-	-
50	0.184	0.176	0.200	0.184	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for precast driven piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.900	9.600	0.675	0.900	-	-	-	-	-	-
10	1.125	14.400	1.350	0.450	-	-	-	-	-	-
15	1.800	9.600	1.575	0.900	-	-	-	-	-	-
20	28.800	27.200	27.200	28.800	-	-	-	-	-	-
25	1.125	0.900	0.900	0.900	-	-	-	-	-	-
30	32.000	32.000	30.400	28.800	-	-	-	-	-	-
35	33.600	33.600	32.000	22.400	-	-	-	-	-	-
40	33.600	33.600	32.000	22.400	-	-	-	-	-	-
45	33.600	35.200	32.000	32.000	-	-	-	-	-	-
50	36.800	35.200	40.000	36.800	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.062	0.031	0.048	0.062	-	-	-	-	-	-
10	0.077	0.047	0.090	0.032	-	-	-	-	-	-
15	0.113	0.031	0.101	0.062	-	-	-	-	-	-
20	0.094	0.088	0.088	0.094	-	-	-	-	-	-
25	0.077	0.062	0.062	0.062	-	-	-	-	-	-
30	0.104	0.104	0.099	0.094	-	-	-	-	-	-
35	0.109	0.109	0.104	0.073	-	-	-	-	-	-
40	0.109	0.109	0.104	0.073	-	-	-	-	-	-
45	0.109	0.114	0.104	0.104	-	-	-	-	-	-
50	0.120	0.114	0.130	0.120	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.405	4.320	0.304	0.405	-	-	-	-	-	-
10	0.506	6.480	0.608	0.203	-	-	-	-	-	-
15	0.810	4.320	0.709	0.405	-	-	-	-	-	-
20	12.960	12.240	12.240	12.960	-	-	-	-	-	-
25	0.506	0.405	0.405	0.405	-	-	-	-	-	-
30	14.400	14.400	13.680	12.960	-	-	-	-	-	-
35	15.120	15.120	14.400	10.080	-	-	-	-	-	-
40	15.120	15.120	14.400	10.080	-	-	-	-	-	-
45	15.120	15.840	14.400	14.400	-	-	-	-	-	-
50	16.560	15.840	18.000	16.560	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the construction of the proposed six storied residential building.

Soil profile scenario

Overall soil information of the investigated site is quite regular in between the boreholes. However, vertical inconsistency exists in the prevailing soil formation. In all the boreholes clay soil of soft to very soft consistency has been observed up to 15 ft depth followed by dense sandy layer of about 5 ft thickness. Then again, a very soft clay has been encountered at the 25 ft depth which is followed be dense sand deposition up to 50 ft depth.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed structure. Since the whole site is dominated by cohesive soil with soft to very soft consistency in the top 25 ft depth with an intermediate single sandy layer, shallow foundation without ground improvement is not suitable for this type of structure. In this regard, the whole site in accordance with the borehole condition may be densified by sand drain or wick drain method. The depth of ground improvement should be selected by the foundation/design engineer considering both pressure influence zone and degree of densification required to minimize uneven settlement. After soil improvement, appropriate pattern of shallow foundation may be selected on the basis of degree of ground improvement. Although, bearing capacity values for designing shallow foundation can be found from the attached table (7) in the report which is based on existing soil condition; it is necessary to reinvestigate the modified bearing capacity of the densified stratum after ground improvement. However, to overcome the ground improvement difficulties deep foundation may be adopted.

Safety provisions

Since al the boreholes of the proposed site are dominated with cohesive soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the settlement of the site considering building load while designing either shallow or deep foundation. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. No further investigation was done by the client.

Special notes

These suggestions are absolutely tentative, and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50 ft** as client provided.

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